Capacity and departure scheduling in MATSim

LSGI3213 Intelligent Transportation Systems

Group leader: ZHANG Shizhuo

20-minute presentation

Research questions

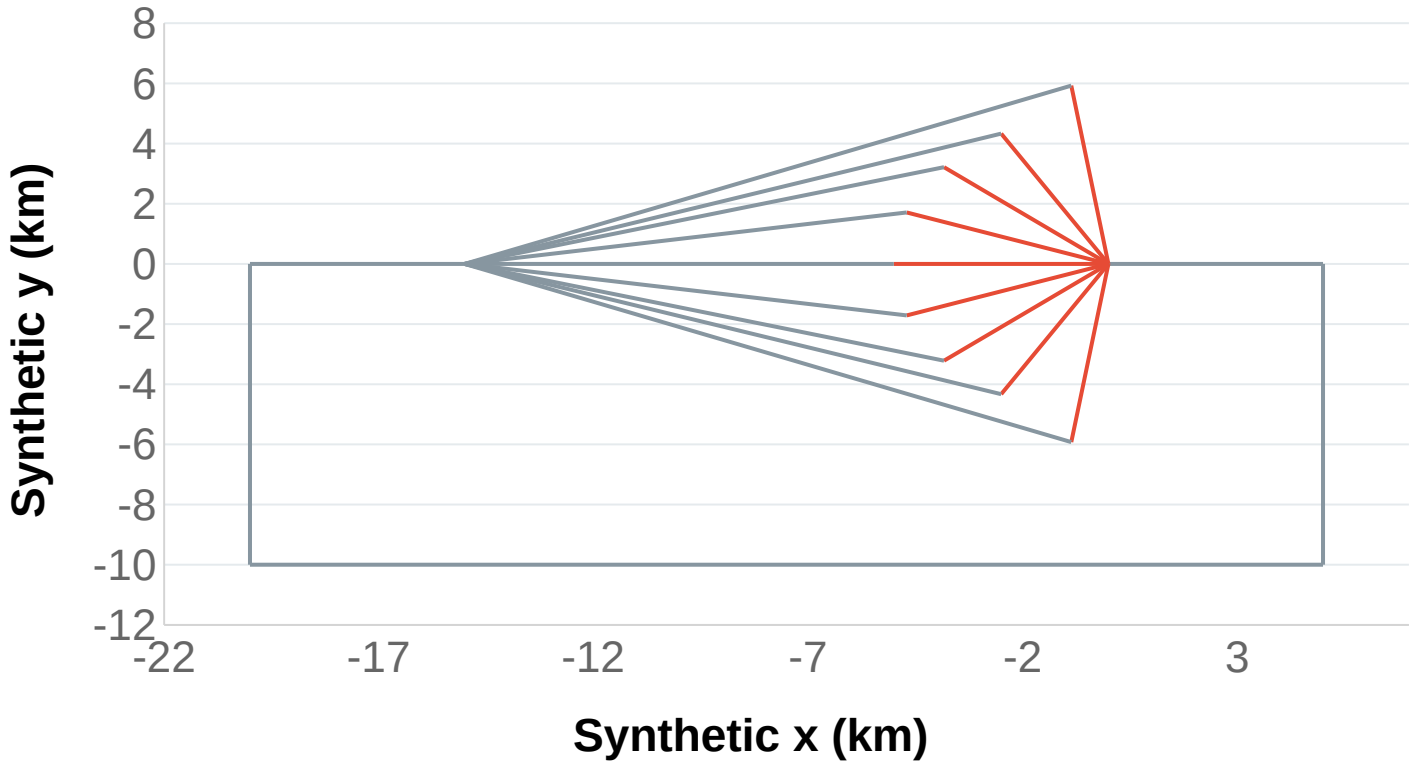
What changes when bottleneck capacity doubles?

What changes when the same travellers leave across one hour?

Do capacity and scheduling provide overlapping benefits?

How does learned departure timing differ from an assigned schedule?

Official equil data



15 nodes; 23 directed links

2,000 agents: home-work-home

Nine outward corridors; shared return path

Downloaded inputs match the tutorial files byte for byte

Source: MATSim matsim-12.0 equil example; public repository and SHA-256 manifest included

Course reference and congestion benchmark

Course reference: unchanged 100-person plans; 201 trips

Congestion benchmark: 2,000 people; 4,000 trips

Policy effects compare the same 2,000 people

Course ZIP: all four input files verified byte for byte

Project report; saved MATSim results

MATSim execution and learning

Select a remembered daily plan

Execute all agents together in QSim

Score activities, travel and timing

ReRoute explores alternatives; BestScore selects remembered plans

Sources: MATSim Book; MATSim 12.0 configuration and Java source

Five controlled scenarios

Scenario	Supply / schedule	Replanning
Baseline	All first departures at 06:00	BestScore 0.9; ReRoute 0.1
Capacity	Links 11-19: 1,000 → 2,000 veh/h	Same
Stagger	Departures spread 05:30-06:30	Same
Combined	Capacity + Stagger	Same
Adaptive	Baseline initial network and plans	0.1 time mutation; BestScore 0.8

Common controls: scoring, demand, facilities, free speed, lanes, iterations and seed set

Execution and independent checks

15 main runs: 5 scenarios × 3 seeds

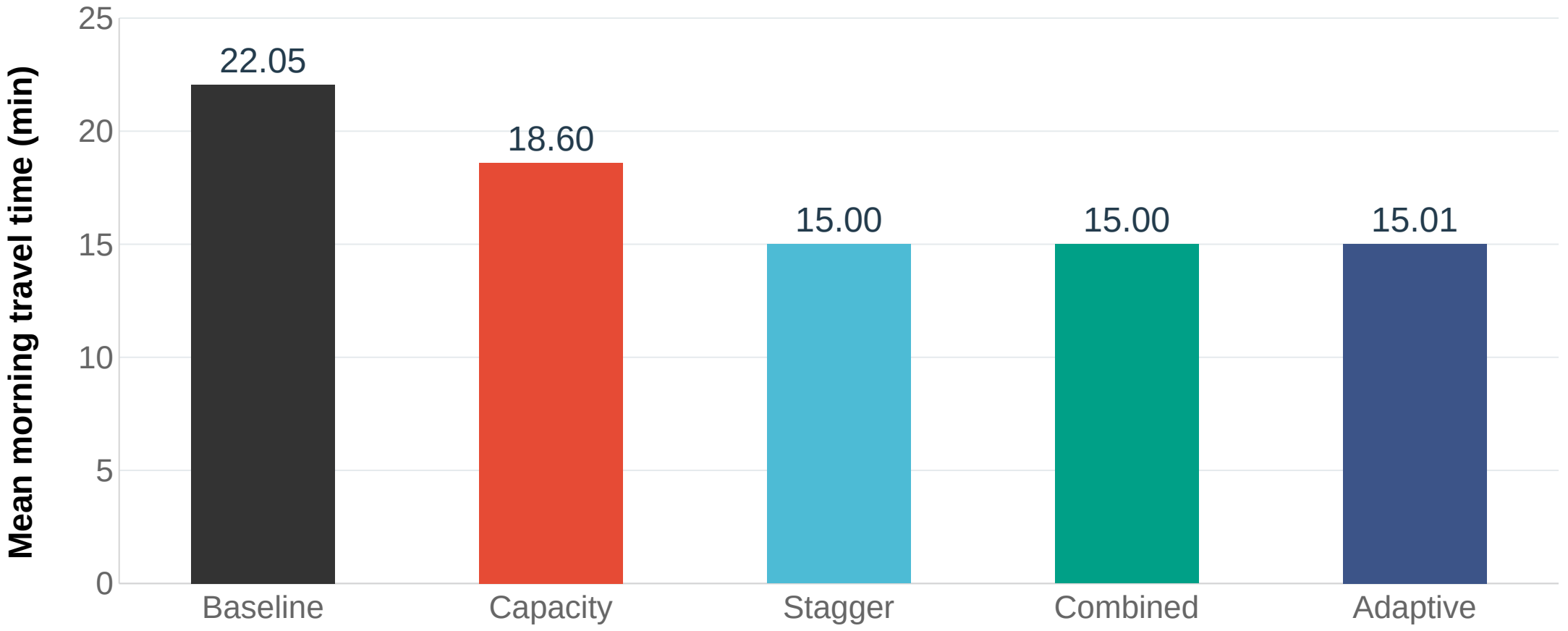
Iterations 0-100; seeds 4711, 42 and 2025

4,000 completed trips per main run; zero stuck agents

Event-based trip times agree with MATSim native trip tables

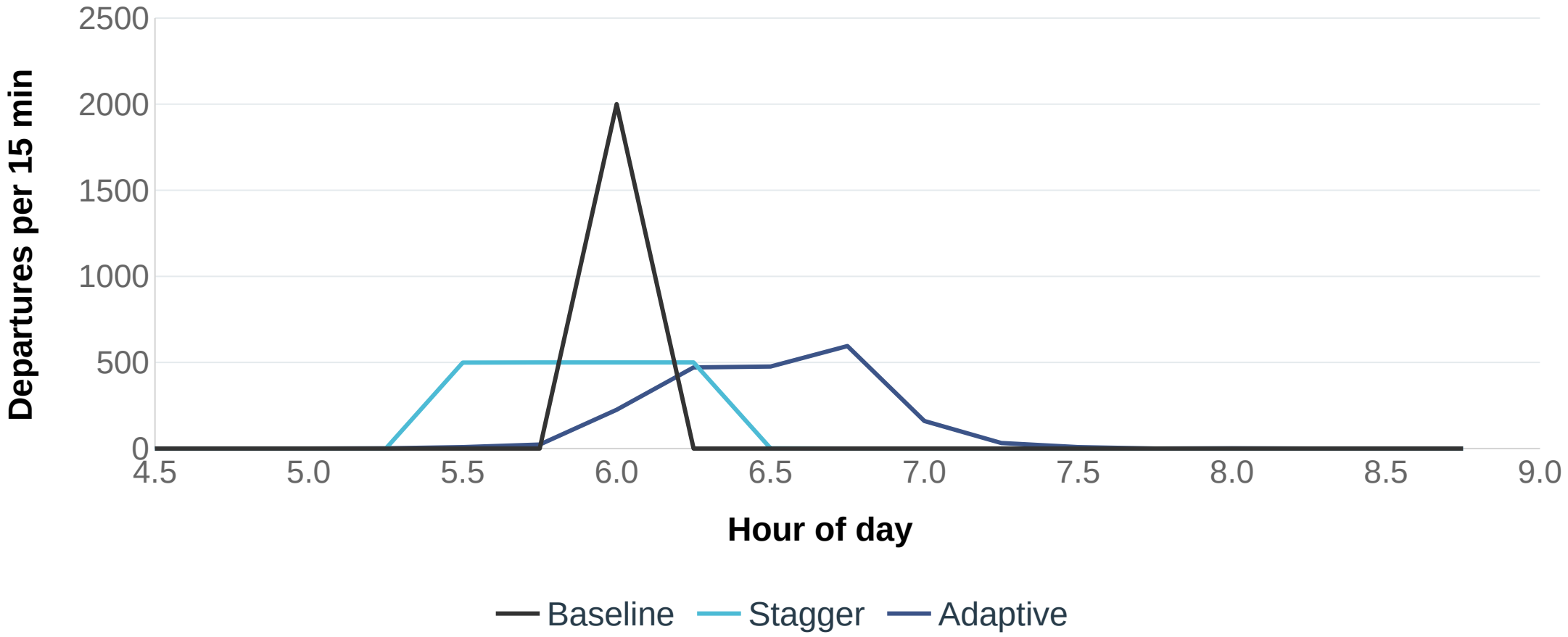
MATSim 12.0; Java 11.0.30; saved XML, event files, logs and validation code

Morning travel time



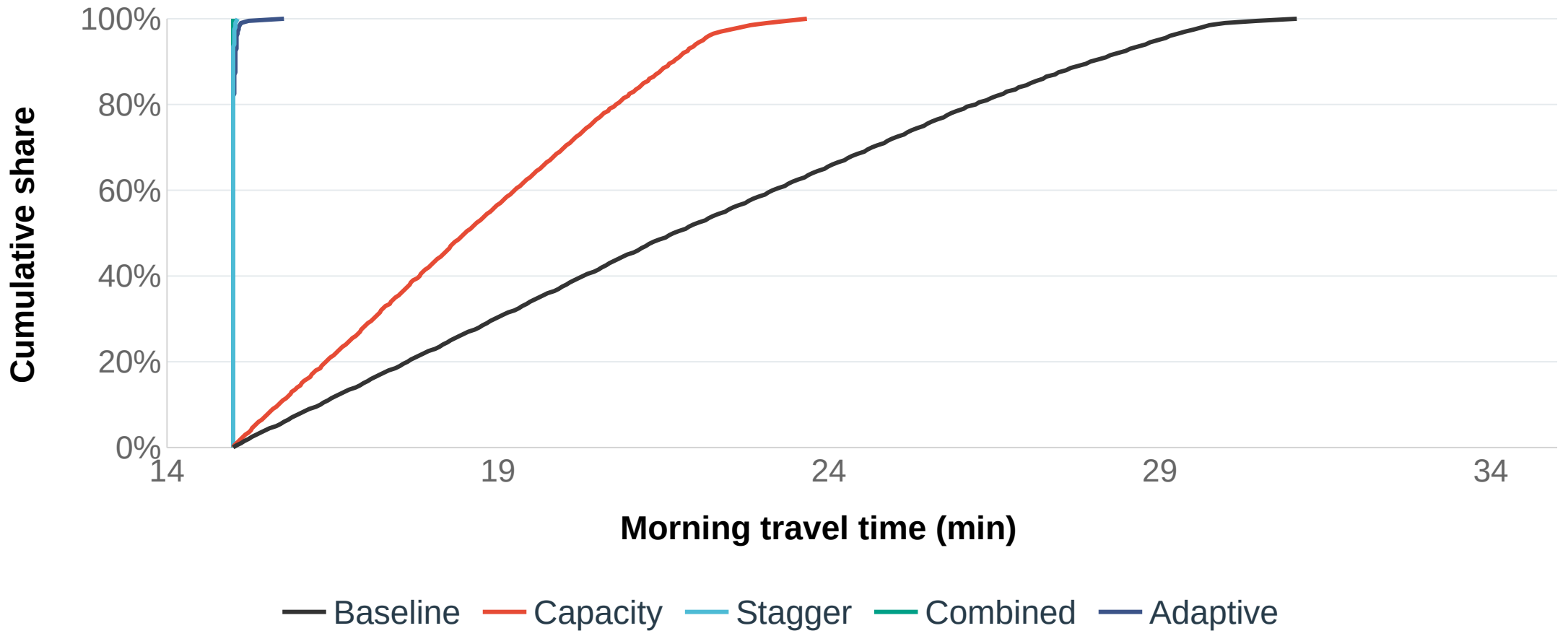
Mean of three final-iteration runs; 2,000 morning trips per run; bar axis starts at zero

Realised departure patterns



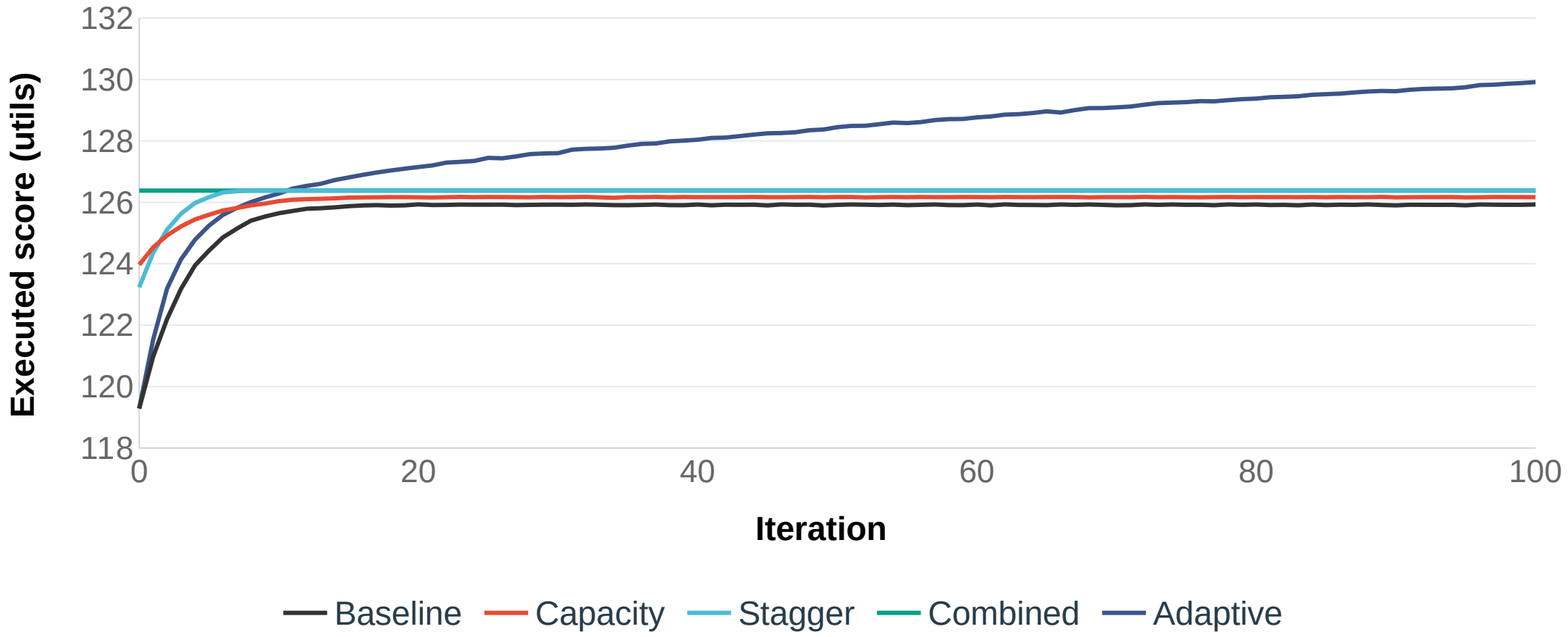
Seed 4711; matching 5-minute histogram is supplied in the report and figure data

Travel-time distributions



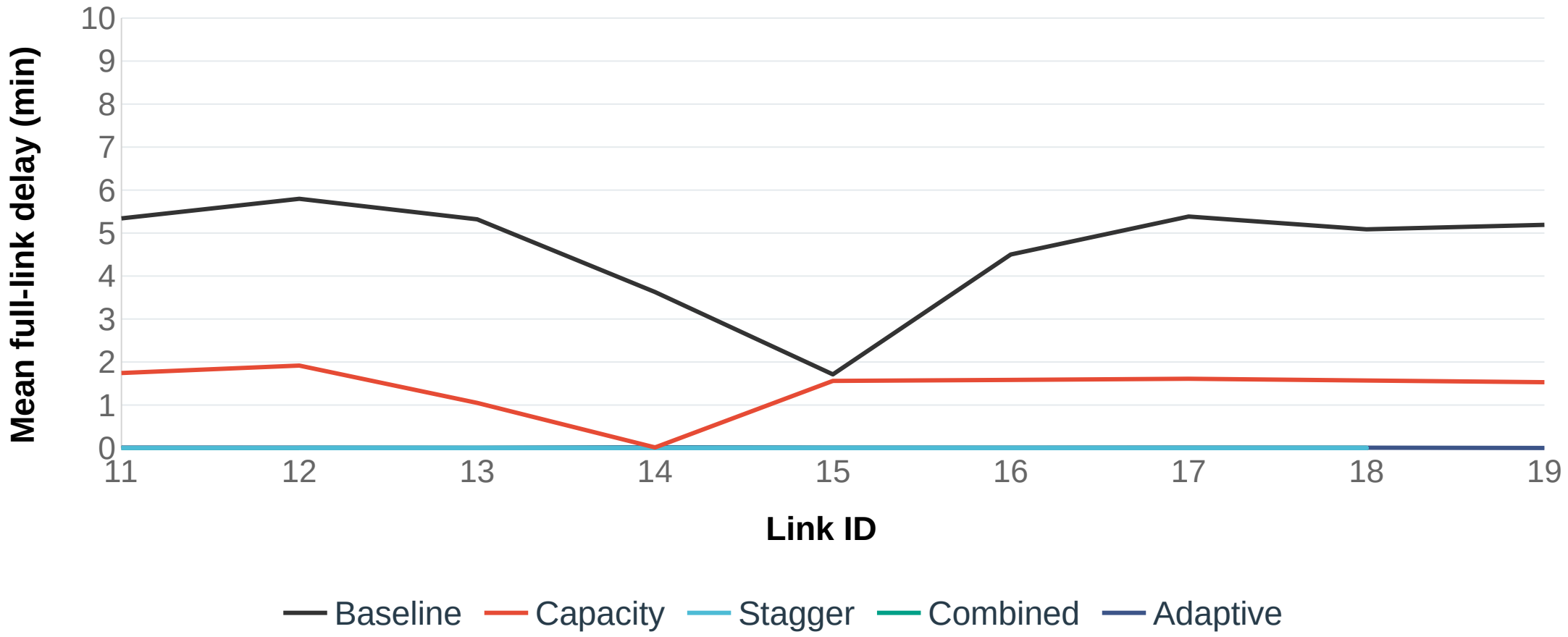
Seed 4711; 2,000 morning trips per scenario; ordered points sampled for display

Executed scores and timing preferences



Seed 4711 shown; identical scoring rules; three-seed histories and extended runs in report

Delay on the nine corridor links



Seed 4711; delay = max(traversal time − length/free speed, 0); endpoint links excluded

Robustness and convergence limits

Scenario	AM mean: three-seed range	Tail score change
Baseline	21.935 - 22.145 min	-0.0003 utils
Capacity	18.555 - 18.632 min	-0.0011 utils
Stagger	15.002 - 15.002 min	-0.0000 utils
Combined	15.000 - 15.000 min	+0.0000 utils
Adaptive	15.010 - 15.012 min	+0.2451 utils

Tail change: mean score at iterations 91-100 minus 81-90; 200-iteration checks in report

Interpretation for ITS

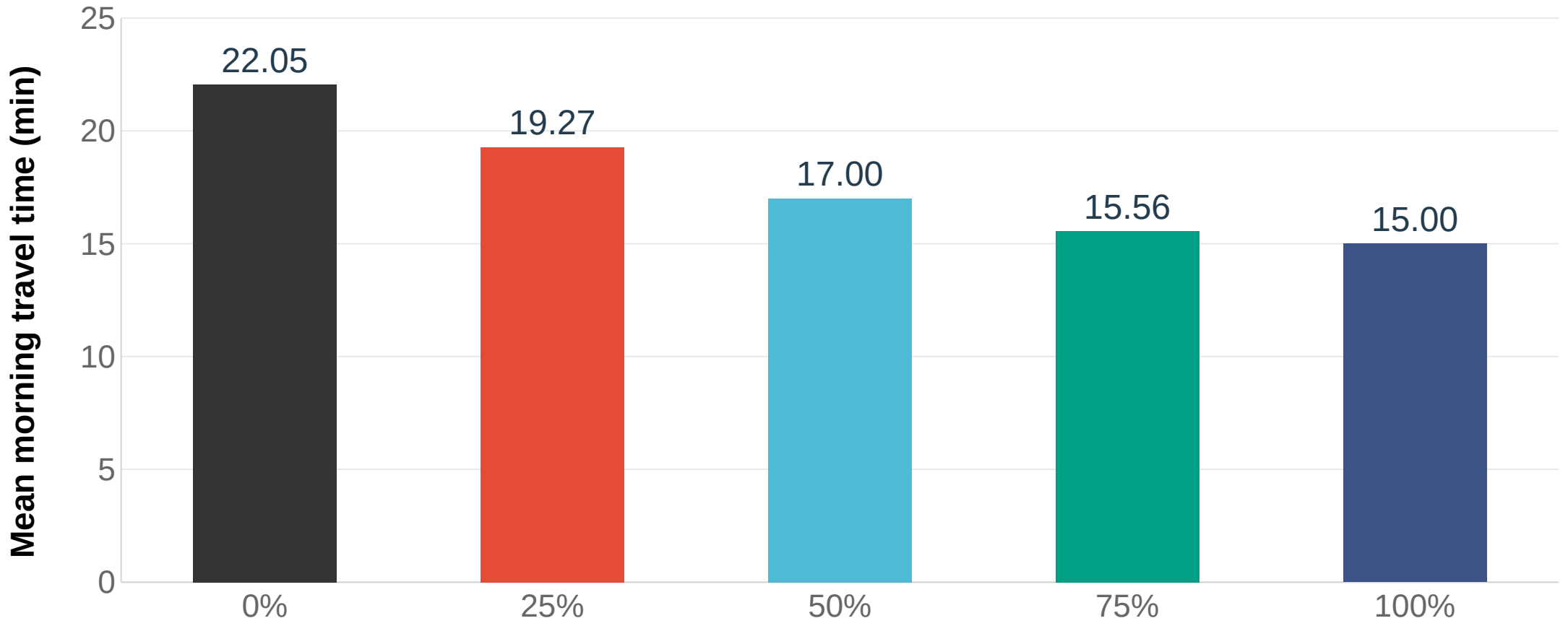
Capacity and scheduling overlap: interaction = +3.45 min

Staggering assumes full participation and flexible schedules

Capacity change raises discharge flow; it does not add lanes

Stagger / Combined: 41.2-41.6% have lower daily scores

Sensitivity to participation



Three seeds per rate; fixed participant selection; 0% / 100% reuse Baseline / Stagger

Why the benefits overlap

Baseline: 22.05 min; Capacity: 18.60 min

Stagger: 15.00 min; Combined: 15.00 min

Both interventions remove part of the same delay

No cost model: time savings cannot establish value for money

Project report; saved MATSim results

Average improvements and individual losses

Stagger / Combined: 41.2-41.6% have lower daily scores

Scores reflect travel and activity timing

Pair the same person IDs within each simulation seed

No demographics: no claim about income or neighbourhood equity

Project report; saved MATSim results

Conclusions

Baseline morning duration: 22.05 min

Capacity saving: 15.7%; Stagger saving: 32.0%

Combined duration: 15.00 min; extra capacity adds little once queues are removed

Interpret travel, daily scores and link states together

Sources and acknowledgment

MATSim developers: official equil inputs, matsim-12.0 tag

Horni, Nagel & Axhausen (2016), The Multi-Agent Transport Simulation MATSim

OpenAI Codex assisted with data discovery, code, execution, analysis, report and slides

Full references, source hashes and AI disclosure accompany the report

github.com/matsim-org/matsim-libs | doi.org/10.5334/baw